

STARS Adaptation Precipitation Depictions (U.S. Air Traffic Control)

Seed Log (Active Seed – SL-001)

Timeframe: Recorded: Winter 2026; March 2026

Revision Note: Formatting and timeframe standardization updated (Version 1.1)

Observation: Three Certified Professional Controllers (CPCs), including the operator, were unsure about the precipitation intensity levels depicted on the radar scope.

Trigger: The operator recalled that several months had passed since the new Standard Terminal Automation Replacement System (STARS) adaptation (a unique configuration of the radar scope) had been implemented. Despite this, none of the CPCs present could immediately distinguish which depiction represented moderate precipitation and which represented heavy precipitation and had to consult a posted reference near the radar scope.

Note: The original Seed Log was recorded in a physical notebook. The entry presented here is a reconstructed Seed Log created during preparation of this example.

V.A.I.Y.L. Sequence

Variation

CPCs are expected to quickly distinguish between the six precipitation intensity levels displayed on the radar scope with little effort. In this case, the operators needed to consult the posted reference near the radar scope to determine which depiction represented moderate and heavy precipitation.

Analysis

The operator noted that under the previous STARS adaptation, heavy precipitation had been depicted as a solid yellow-orange color without texture. Under the new adaptation, the same presentation now represents moderate precipitation.

Examination (external to the framework)

Notes: N/A

Operational Process: N/A

Operational Review: N/A

Insight

The new STARS adaptation effectively changed how precipitation intensities are visually interpreted by CPCs at the facility. Although operators had likely been briefed on the change, the yellow-orange color may still be subconsciously associated with heavy precipitation.

Yield

Three CPCs clarified their understanding of the precipitation depictions introduced by the new STARS adaptation.

Learn

The operator became more aware of the potential confusion the new STARS adaptation may have introduced at the facility.

Scope

Uncertain. If the STARS adaptation is unique to the facility, the confusion between precipitation intensities may be limited to the facility. If similar adaptations were implemented elsewhere, the issue may exist at a broader system level.